Contents

I Synthesis **2**

1 What the project established, refuted, and left open **3**

1.1 Claims and their strength 3

1.2 What we got out of it 4

2 What worked and what did not **5**

2.1 Demotivations, and why they mattered 5

2.2 Lessons 6

3 Decision log **7**

4 Validity, recommendations, and next steps **9**

4.1 Threats to validity 9

4.2 Recommendations 10

4.3 Publication options and open decisions 10

PART I

Synthesis

1

What the project established, refuted, and left open

1.1 Claims and their strength

Table 1.1: Every substantive claim of the project and how strongly the evidence supports it.

Claim	Strength	Evidence and boundary
Public cytology benchmarks hide a crop-to-parent hierarchy	established	Study 1: recovered and verified for four datasets; Herlev negative control at zero.
Random crop splits share parents and pixels	established	92.6% (SIPaKMeD), 99.97% (CRIC) of test cells with a training parent; 52.6% / 71.5% / 34.0% with shared pixels.
The parent is recognisable	established	Study 1 probe 109× chance; Study 2 encoder 102× chance.
Crop-level splitting inflates accuracy	established (registered, powered)	H1 +0.97 pp, all twelve architectures; H2 +12.08 pp; five inference methods agree.
The inflation is leakage, not diversity	established (registered, replicated)	H3 +1.73 pp, diversity 0.00 pp; replicated on fresh seeds (+1.47 pp).
Exposure follows structure	supported	CRIC, with 29 cells per slide and high class–slide dependence, inflates ten times more than SIPaKMeD; predicted by the closed forms.
The leak needs no background	established (registered)	F3 +1.86 pp with every background masked; test-time removal leaves 68%.
The leak is tied to the field's colour rendering	supported (registered), qualified	F2 removes 74% at test time; F4 shows colour jitter in training does not prevent it.
A background context route carries the leak	not supported	H4 0.42 pp (−0.35 to 1.18).
Models apply a slide-level label prior (CRIC)	refuted	F1 reversed (−4.29 pp); equal gains for discordant and concordant cells.
The leak is memorised shared pixels	not supported (exploratory)	Equal advantage on non-overlapping crops.
The inflation is carried by linearly decodable field identity	refuted within its scope	F5, F7, F8: recoverability cut 38–91%, gap flat (pooled slope 0.000, CI −0.005 to 0.005).
Saliency maps reveal the leak	not supported (descriptive)	Grad-CAM identical for leak and clean models.
Stronger grouping (P3) is more conservative than P2	not supported	P2 − P3 0.03 pp; the ordering passes for 4 of 12 architectures.

1.2 What we got out of it

Scientific results

A certified, powered estimate of how much crop-level splitting inflates two public cytology benchmarks; a causal separation of leakage from diversity; a mechanism narrowed by elimination; and a methodological warning that the standard audit tools (probes, saliency, augmentation) cannot detect or remove this kind of leakage.

Reusable artifacts

Verified crop-to-parent maps for SIPaKMeD, Cx22, CRIC, and Herlev; registered P1/P1′/P2/P3 splits with hashes for 20 seeds; the 2C factorial designs and transplant plans; a risk calculator (`cytofair risk-report`) that predicts a benchmark's exposure from its parent table; a deterministic GPU-resident training stack with complete per-run artifacts; a fail-closed scheduler and collectors; exported records of every run (`results/study2/run_records`); 121 GB of archived predictions, features, and weights for future analyses without retraining.

Writing

A single-column manuscript (`paper/v2`, 17 pages, every number generated), the earlier ISBI draft, the Study 1 record, and this lifecycle record.

2

What worked and what did not

Table 2.1: What worked and what did not, by area.

Area	Worked	Did not work, or cost more than expected
Question design	Separating structure, availability, consequence, and mechanism kept every claim at its true strength.	Study 1 asked the consequence question six times and could not answer it.
Statistics	One crossed mixed model with a split random effect, sized by simulation; registered sensitivity analyses that all agreed; the assumption trigger worked as designed.	Per-architecture testing in Study 1; Satterthwaite *df* for H3 is only 3 with four architectures, so its interval is wide.
Causal designs	The 2C factorial cleanly separated leakage from diversity; evaluation-only follow-ups on saved models were cheap and decisive.	The 2B transplant contrast was noisy (sibling backgrounds help even clean models); the 2A MLP adversary was evaded by ConvNeXt-T.
Mechanism	Elimination with several independent manipulations gave a coherent picture.	No single test identifies the carrier positively; F2 and F4 pull in different directions.
Engineering	GPU-resident augmentation (exact torchvision port); packed hashed caches; bitwise determinism; complete per-run artifacts; zero failed tasks in 2,921.	The four-core CPU quota had to be discovered the hard way; the slow network home forced a strict storage discipline.
Governance	Register, smoke-test on non-contrast splits, dry-run, collect fail-closed, amend with dates; every deviation recorded.	A dry run that bypassed one pipeline stage let a labelling bug through to the first real analysis (Amendment 9). Registrations are local commits, not yet externally timestamped.
Operations	Detached controllers, status files, sequential offloading with verification, clean vacating of node-local storage.	Several shell and remote-wait mistakes (self-matching process searches, hung SSH sessions) cost time but no data.

2.1 Demotivations, and why they mattered

Four results were disappointing at the time. Study 1's six tests failed to survive correction, which looked like a null; Study 2 showed it was underpowered. H4 and H5 failed to isolate a mechanism; Amendment 8 showed why (the leak is neither in the background nor in a linearly readable code).

F1 went the opposite way from the prediction; that reversal is itself evidence against a slide-level label prior. And the ConvNeXt-T adversary was evaded; that failure is a concrete example of why decodability and use must be measured separately. Each setback narrowed the explanation instead of ending it, and none was hidden.

2.2 Lessons

1. Ask a pooled question once, with the dependence structure in the model (the shared test set per seed is a random effect, not twelve independent tests).
2. Keep every per-sample artifact. Half of Amendment 8 cost no training because Study 2 kept logits, features, and weights.
3. Measure manipulations independently of the mechanism they target: the gradient-reversal adversary's own accuracy fell while an independent probe's rose.
4. Dry runs must pass through every stage of the real pipeline.
5. On shared hosts, discover the real resource limits (CPU quota, writable disks, network storage behaviour) before designing the execution.

3

Decision log

Every design or execution decision that shaped a result, with the reason it was taken and the alternative it displaced.

Table 3.1: Decisions and their reasons.

Decision	Reason	Alternative not taken
Study 2 statistics		
Add a split random effect to H1	Twelve architectures share each seed's test set; without it the model treats twelve evaluations of one test set as independent.	The roadmap's formula without it.
Twenty seeds	Smallest count meeting the registered power rule (0.90 at pilot variances, 0.75 at 1.5×).	15 seeds (failed the inflated criterion); 25–30 (more compute for little power).
Twelve architectures	Breadth across CNNs, hierarchical transformers, plain ViTs, and a self-supervised model, so the effect is not tied to one family.	The six Study 1 models only.
Bootstrap primary when residuals are heavy-tailed	Registered before data; protects the primary test from non-normal seed-level noise.	Always the Satterthwaite t-test.
H2–H5 in one Holm family; H1 alone	H1 is the question; the mechanism tests are a family of related but separate claims.	One family of five (would have penalised the primary).
Study 2 designs		
220 crops per class in 2C	Largest budget feasible for every registered seed and 100/100 unrelated seeds; 240 failed for seed 307.	240 or 260 crops.
Common test set across all four 2C cells	Every cell is scored on the same crops, so differences are training effects only.	Separate test sets per cell.
Diversity estimated within clean cells only	The low-diversity leak cell gives each test crop more siblings, so the interaction confounds dose with diversity.	A full 2×2 interpretation.
Sibling background as the 2B control	Isolates the cell's own acquisition context from class-typical context.	Only unseen-field backgrounds.
Class-conditional parent adversary	Suppressing parent identity must never require removing class information.	An unconditional 424-way adversary.
CRIC as 100×100 windows	CRIC releases nucleus coordinates on slide images, not crops; fixed windows are the simplest faithful crops.	Variable-size crops from annotations.
Execution		
GPU-resident augmentation	Four-core CPU quota starved data loaders; an exact port keeps the registered recipe.	PIL augmentation in DataLoader workers.

Decision	Reason	Alternative not taken
One training job per GPU	GPUs were saturated by one job; two jobs shared throughput without speed-up.	Two jobs per GPU.
Math attention kernel only	Fused attention kernels have non-deterministic backward passes.	Faster fused kernels.
Float16 checkpoints	Halves storage of about 117 GB of weights; re-evaluation reproduces 418/480 models exactly, the rest within three crops.	Float32 checkpoints.
Amendment 7 before any 2A run	Deterministic divergence would otherwise have failed the whole arm.	Handling divergence after seeing it.
Registered analysis run on the GPU host	Run records point at node-local artifacts; moving gigabytes to a workstation was slower than computing in place.	Pulling all artifacts to the workstation.
Amendment 8		
A separately registered follow-up	Study 2's mechanism tests were inconclusive, and saved artifacts made decisive tests cheap; registering first keeps them confirmatory for new questions.	Unregistered exploration, or re-testing H4/H5 with more seeds.
Reuse saved models where possible	F2 and F5 needed no training; three arms cost 7 GPU-hours in total.	Retraining for every arm.
Fresh seeds for retraining arms	New data for new questions; no reuse of Study 2 mechanism seeds.	Reusing the mechanism seeds.
Probe-matched linear adversary	ConvNeXt-T had evaded the MLP adversary; a standardised linear head cannot hide identity from a linear probe while leaving it readable.	Re-running the MLP adversary.
LEACE ranks up to 256	The smallest feature dimension is 384 (DINOv2 ViT-S); 256 removes most within-class parent structure in every backbone.	Full-rank erasure (exceeds some feature dimensions).
HED-strong colour augmentation	A published, strong, stain-shaped perturbation.	Hue/saturation jitter only.
One device probe for every arm	Recoverability must be measured identically across manipulations; agreement with sklearn verified ($r = 0.999$).	Mixing probe implementations.
Pixel-overlap split labelled exploratory	Added after the registered results to test a specific alternative explanation.	Presenting it as confirmatory.
Close-out		
Offload checkpoints during the follow-up matrix	52 GB of new checkpoints against a 71 GB margin; verified copies before deletion.	Pausing the matrix on the disk guard.
Recommend ISBI framing around audit tools	The registered evidence is strong; the novel take-home for an imaging audience is that probes and saliency cannot detect this leakage.	Framing around "leakage exists".

4

Validity, recommendations, and next steps

4.1 Threats to validity

Parents are not patients. SIPaKMeD fields, Cx22 source images, and CRIC slide images are the deepest released units. Several can come from one patient, so parent-disjoint evaluation is necessary but may not be sufficient for clinical independence; every inflation estimate is a lower bound on what patient-level grouping would remove.

One recipe. Every model used one fixed fine-tuning recipe. Stronger augmentation, longer training, or other optimisers could change the size of the inflation and the models' reliance on acquisition context.

Smaller training sets in the factorial. Matching crop and field counts required 220 crops per class (1,100 in total against 3,239 in Phase 1); the 2C, 2B, and carrier effects are measured on smaller training sets than the headline contrast, and the decomposition that transfers them is approximate.

Imperfect interventions. Foreground masks are threshold rules; inpainting leaves smoothed backgrounds; Reinhard transfer matches only first and second moments; erasure and adversaries act on pooled features and linear structure.

Post-results follow-up. Amendment 8 was motivated by Study 2's results. Its hypotheses and analyses were fixed before any follow-up run, but they are follow-up questions; the pixel-overlap split was added afterwards and is exploratory.

CRIC crops. The CRIC cells are windows cut by this project around published nucleus coordinates, not an official crop release, so absolute CRIC accuracies are not directly comparable with published numbers.

External verification. The registration commits have not yet been pushed or deposited; until they are, their dates rest on the repository and on the commit hashes recorded inside every run.

No clinical claim. The project audits research benchmarks; it does not estimate screening sensitivity, specificity, prevalence-calibrated performance, or effects on readers.

4.2 Recommendations

1. **Release the hierarchy.** Every observation should carry its acquisition parent (field, source image, or slide), a patient or specimen identifier where permitted, and, if audited, a duplicate-component identifier.
2. **Split at the deepest released unit and publish the split**, with its hash.
3. **Report both units**: observation and parent counts per split, and the fraction of test observations with a training sibling.
4. **Check the structure before training** with the closed forms (`cytofair risk-report`).
5. **Treat one question as one test**, with a mixed model that treats architectures and splits as random effects, sized by simulation.
6. **Do not certify a split with a probe, a saliency map, or an augmentation.** Leakage inflated accuracy with field identity erased from the features, with identical Grad-CAM maps, with backgrounds removed, and with strong colour augmentation.
7. **Keep the evidence**: per-sample predictions, final weights, and the exact split for every run.

4.3 Publication options and open decisions

4.3.1 Venues

ISBI 2027. Four pages including figures and tables (references may use a paid fifth page); deadline Monday 26 October 2026, 23:59 US Eastern; single-blind; arXiv preprints allowed; a "Compliance with Ethical Standards" section and a disclosure of AI-generated content in the acknowledgments are required; presentation in person in Lausanne. The evidence is stronger than typical for the venue. The risks are perceived novelty (patient-level splitting is textbook advice) and the page limit. The recommended framing is "how much crop-level splitting inflates public cytology benchmarks, and why the usual audit tools cannot see it": H1–H3 as the result, one figure (per-architecture SIPaKMeD with CRIC), one table of all thirteen registered tests, and a paragraph on F2, F3, and the probe and Grad-CAM findings. Everything else waits for the journal version.

Journal version. The full manuscript (`paper/v2`) with Study 2 and Amendment 8. Check the journal's policy on extended conference papers if ISBI is pursued first; ISBI only forbids simultaneous submission of a substantially similar paper to another conference or workshop.

Preprint. Allowed by ISBI; it would also give the registrations a public timestamp once the repository is pushed.

4.3.2 Decisions only the authors can make

- Push the `study2` branch (and the Cx22 commits) to the public repository, and deposit the preregistrations with a third party (for example OSF) for an external timestamp.

- Choose a software licence; confirm redistribution terms for derived artifacts (SIPaKMeD, CCEDD).
- Co-author approval of every posted or submitted version; fill the e-mail and repository placeholders.
- Whether to submit to ISBI 2027, and who presents in person.
- The AI-use disclosure wording: much of the Study 2 code, analysis, manuscript text, and this record was drafted with an AI assistant under the authors' direction.
- Whether to open Phase 3 (other imaging domains) and the direct test of the F2–F4 tension.

4.3.3 Future work

- Apply test-time stain transfer to the colour-augmented carrier models to resolve the F2–F4 tension directly.
- Non-linear and spatial (pre-pooling) erasure of field identity.
- Patient-level grouping where a dataset's provenance allows it.
- Other recipes (stronger augmentation, longer schedules, stain normalisation as standard practice).
- Other domains where crops share acquisition units (the pre-specified fish-imagery V2).

Bibliography